

	0 1 2 3 4 5 6 t/s
7(a)(i)	Binding energy of a nucleus is the minimum energy required to break the nucleus into its individual nucleons. It is the energy equivalent of mass defect and is calculated by product of the mass defect and square of the speed of light.
(a)(ii)	Mass defect = $11 \times 1.00814\text{u} + 12 \times 1.00898\text{u} - 22.99706\text{u} = 0.20024\text{u}$ Binding energy per nucleon = $\frac{0.20024 \times 931.4}{23} = 8.109 \text{ MeV}$
(b)(i)	Mass change = $22.99706\text{u} + 1.00898\text{u} - 23.99857\text{u} = 7.47 \times 10^{-3} \text{ u}$

(b)(ii)	Energy released = $7.47 \times 10^{-3} \times 931.4 \text{ MeV}$ $= 7.47 \times 10^{-3} \times 931.4 \times 1.6 \times 10^{-3}$ $= 1.113 \times 10^{-12} \text{ J}$ $\Rightarrow \frac{hc}{\lambda} = 1.113 \times 10^{-12}$ $\Rightarrow \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{\lambda} = 1.113 \times 10^{-12}$ $\Rightarrow \lambda = 1.80 \times 10^{-13} \text{ m}$
(c)	Using $A = A_0 e^{-\lambda t}$ $\frac{1}{20} A_0 = A_0 e^{-\lambda(65)}$ $\frac{1}{20} = e^{-\lambda(65)}$ $\ln \frac{1}{20} = -\lambda(65) \ln e$ $\lambda = 0.046 \text{ hour}^{-1}$
8(a)(i)	The relative speed of approach of the two objects is greater than their relative speed